

Moamen Moustafa Basyoni

Software Engineer | Python APIs, Cloud & AI Integration · PropTech

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PROFESSIONAL SUMMARY

Engineer with around three years building API-driven, distributed backends and the automation on top of them. Python services on FastAPI and Flask with async/await, Pydantic v2, SQLAlchemy 2.0 and pytest over PostgreSQL; distributed architecture with event-driven messaging, service discovery and an API gateway. Cloud across both major providers — Azure managed services and AWS EC2/S3 — including a migration driven by cost. Builds AI into products rather than alongside them: RAG and multimodal search, agentic workflows, and AI tooling that automates a full delivery pipeline. Previously the technical partner on a **short-term rental / PropTech platform**, built from scratch.

PROFESSIONAL EXPERIENCE

R&D Engineer — Backend, Cloud & AI · Andalusia Health Business Solution (AHBS)

Dec 2023 – Present · Alexandria, Egypt

- Built **Python** services with **FastAPI** (Pydantic v2, async/await, dependency injection) and Flask, with **SQLAlchemy 2.0** and Alembic migrations over **PostgreSQL** and SQL Server; async HTTP with asyncio and httpx; tested with pytest and pytest-asyncio.
- Designed **API-driven architecture** across distributed services — REST and gRPC contracts authored before implementation, improving 10+ inter-service interaction patterns.
- Built distributed infrastructure: **Temporal Saga orchestration** for long-running transactions, **RabbitMQ** event-driven messaging, Consul service discovery, and an Nginx API gateway with health-aware load balancing — cutting deployment downtime from 2 hours to 15 minutes.
- **Automation at scale**: built an AI-driven delivery pipeline covering requirements analysis, story generation and quality scoring, test-case and test-plan creation, Playwright-driven test automation, and automated pull-request review that comments line-by-line and checks implementations against their business requirements.
- Built **five custom Model Context Protocol servers** (Azure DevOps, Figma, XMind, Playwright, database) and a code graph spanning **38 modules**, surfaced through a custom Azure DevOps extension.
- **AI integration in products**: RAG and GraphRAG retrieval on Neo4j, multi-model orchestration across OpenAI and OpenRouter with cost controls and observability, and shipped assistants for mail, assessment, and booking.
- Implemented an OpenID Connect / OAuth2 identity server with JWT and scope-based authorization; OWASP-aware coding and PII masking.
- CI/CD pipelines in Azure DevOps with Docker across Dev, Staging and Pre-Live; migrated source control from a legacy TFVC monolith to Azure DevOps Git multi-repo.

Technical Partner (equity) — Trastain · PropTech / short-term rentals

Feb 2025 – Aug 2025 · Cairo, Egypt (Remote) · Part-time

- Joined a friend's company as technical partner on an equity stake and **built the short-term rental booking platform from scratch**.
- Ran the competitor analysis across existing short-term rental and Airbnb-style platforms, identified that the Airbnb model is not culturally aligned with the Egyptian and wider Middle Eastern market, and turned that into the product's positioning and feature set.
- Developed **RAG-based property matching** and **multimodal AI search** — query by image, location, or natural-language description.
- Implemented shared-property booking, preference matching, policy enforcement, and transaction security.
- Integrated Stripe and Paymob with webhook processing and HMAC signature verification.
- Shipped via GitHub Actions CI/CD to Azure Web Apps and Azure Static Web Apps.

Freelance / Independent Client Work

Alongside employment · with a small partner group

- **Cloud architecture across Azure and AWS**: chose fully managed Azure services (App Service, Blob, managed SQL, Static Web Apps) to reach MVP fast with no self-managed servers; later ran self-managed **AWS EC2 with S3** and AWS messaging services.
- **Cost optimization**: replaced the cloud SMS service with a local provider, cutting messaging spend.
- Built a training-centre management system with session logging, management approval, instructor payment reporting, and two-way **Google Sheets** and **Google Drive** integrations.

Software Engineer — MYM (Make Your Miracle)

Jun 2023 – Oct 2023 · Giza, Egypt

- Backend systems for e-commerce and **booking platforms** — hotel, gym/fitness, and other reservation systems — delivered end to end.

TECHNICAL SKILLS

- **Languages**: Python, C#, SQL, TypeScript, JavaScript
- **Python**: FastAPI, Flask, SQLAlchemy 2.0, Pydantic v2, Alembic, asyncio, httpx, pytest, pytest-asyncio
- **APIs & Architecture**: REST, gRPC, OpenAPI, API gateway, microservices, distributed systems, event-driven architecture, Clean Architecture, SOLID
- **Distributed Systems**: Temporal, RabbitMQ, MassTransit, Consul
- **Databases**: PostgreSQL, SQL Server, Neo4j, Elasticsearch
- **Cloud**: **AWS** (EC2, S3, messaging services) · **Azure** (App Service, Blob Storage, managed SQL, Static Web Apps, DevOps) · provider evaluation and cloud cost optimization
- **AI Integration**: RAG, GraphRAG, multimodal retrieval, LangGraph, CrewAI, Model Context Protocol (MCP), multi-model orchestration (OpenAI, OpenRouter), LangSmith / LangFuse, Claude Code

- **Front-end:** TypeScript, JavaScript, HTML, CSS, jQuery, Ajax
 - **DevOps:** Docker, GitHub Actions, Azure DevOps Pipelines, Git
 - **Integrations:** Stripe, Paymob, webhooks, HMAC verification, Google Sheets/Drive APIs, Microsoft Graph
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EDUCATION

B.Sc. Computer Engineering — Behera Higher Institute (BHI), Alexandria, Egypt · 2023 Grade: B+ · GPA 3.2 / 4.0

PUBLICATION & LANGUAGES

- *IoT-Enabled E-Prescription Management and Dispensing Machine Monitoring* — IEEE, 2024
- Arabic (Native) · English (B2, Upper Intermediate)